MAT1011 Honors Calculus I

Homework 3

2026 Fall

Show your work. Use only methods and results covered in class. Here $\mathbb{N} = \{1, 2, 3, \ldots\}$.

1. Compulsory Exercises

Question 1.

Find the following limits. Justify (a) and (b) using ε–N language, and (c) using ε–δ language. For (d) and (e), give the calculations only.

(a) $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{1}{k(k+1)}$.

(b) $\lim_{n\to\infty} \frac{n}{a^n}$, where $a > 1$.

(c) $\lim_{x\to 1} \left(\frac{1}{1-x} - \frac{3}{1-x^3} \right)$.

(d) $\lim_{x\to 0} \frac{\sqrt{1+x} + \sqrt{1-x} - 2}{x^2}$.

(e) $\lim_{x\to 0} \frac{\cos(2x) - 4\cos x + 3}{x^4}$.

Question 2.

Let $f : [0, 1] \to \mathbb{R}$ be continuous on $[0, 1]$ and differentiable on $(0, 1)$, with $f(0) = 0$ and $f(1) = 1$.

(a) Prove that there are points $0 < \xi < \eta < 1$ such that $f'(\xi) > 0$, $f'(\eta) > 0$ and

$$\frac{1}{f'(\xi)} + \frac{1}{f'(\eta)} = 2.$$

(b) Prove that there is a point $c \in (0, 1)$ such that

$$(1 - c)f'(c) = f(c).$$

Question 3.

Find all vertical, horizontal and oblique asymptotes of each function on its stated domain.

(a) $f(x) = \sqrt{x^2 + 2x} - x, \quad x \in (-\infty, -2] \cup [0, \infty)$.

(b) $g(x) = \frac{\sqrt{x^2+1}}{x}, \quad x \neq 0$.

(c) $h(x) = \frac{x^3 - x^2 + x - 1}{x^2 - 3x + 2}, \quad x \in \mathbb{R} \setminus \{1, 2\}$.

Question 4.

Let f be continuous on $[a, b]$, where $a < b$, and suppose $f(x) \neq 0$ for every $x \in [a, b]$.

(a) Prove that either $f(x) > 0$ for every $x \in [a, b]$, or $f(x) < 0$ for every $x \in [a, b]$.

(b) Prove that there is a number $m > 0$ such that $|f(x)| \geq m$ for every $x \in [a, b]$. Hence prove that $1/f$ is bounded on $[a, b]$.

(c) If $[a, b]$ is replaced by (a, b), must $1/f$ still be bounded? Give a proof or a counterexample.

Question 5.

Let a and b be real numbers, and define

$$F(x) = \begin{cases} ax + b, & x < 0, \\ \cos x, & x \geq 0. \end{cases}$$

(a) Find all values of a and b for which F is continuous at 0.

(b) Find all values of a and b for which F is differentiable at 0, and compute $F'(0)$.

(c) For the values found in (b), is F' continuous at 0? Does $F''(0)$ exist? Give reasons for your answers.

Question 6.

Compute the following derivatives.

(a) $y'(x)$ for $y(x) = \dfrac{\sin(x^2)}{\sqrt{1 + x^2}}, \quad x \in \mathbb{R}$.

(b) $y'(x)$ for $y(x) = \arcsin\left(\dfrac{2x}{1 + x^2}\right), \quad x \in \mathbb{R}$. Also find all points where y is not differentiable.

(c) $P^{(n)}(x)$ for $P(x) = (1 + x^2)\cos x$, where $n \geq 2$ is an integer.

Question 7.

Let $y = y(x)$ be twice differentiable on $(1/2, 3/2)$ and satisfy

$$x^2 + xy + y^2 = 3, \qquad y(1) = 1.$$

(a) Compute $y'(1)$ and $y''(1)$.

(b) Find the tangent and normal lines to the curve at $(1, 1)$.

(c) Find the linearization of $y(x)$ at $x = 1$ and use it to approximate $y(1.02)$.

Question 8.

Let $f(x) = x^3|x-1|$ for $-1 \le x \le 2$.

(a) Find all critical points of f in $(-1, 2)$.

(b) Find the absolute maximum and minimum values of f on $[-1, 2]$, and all points where these values are attained.

(c) Find all local maximum and local minimum points of f in $(-1, 2)$.

Question 9.

Let I be an open interval.

(a) Let $f : I \to \mathbb{R}$ be differentiable, and let $M \ge 0$ be a constant. Prove that

$$|f'(x)| \le M \quad (x \in I)$$

if and only if

$$|f(x) - f(y)| \le M|x - y| \quad (x, y \in I).$$

(b) Suppose f is differentiable on I and $|f'(x)| \le M$ for all $x \in I$, where $M \ge 0$ is a constant. If $a < b$ are points in I and $f(b) - f(a) = M(b - a)$, prove that

$$f(x) = f(a) + M(x - a) \qquad (a \le x \le b).$$

(c) Let $g : I \to \mathbb{R}$. Suppose there is a constant $C > 0$ such that

$$|g(x) - g(y)| \le C|x - y|^2 \qquad (x, y \in I).$$

Prove that g is constant on I. If $|x - y|^2$ is replaced by $|x - y|$, must g still be constant? Give a proof or a counterexample.

2. Extra Exercises

Question 10.

Let I be an open interval containing c, and let $f : I \to \mathbb{R}$ be differentiable at c. Suppose that $\{x_n\}$ is a sequence of zero points of f, with

$$x_n \in I, \qquad x_n \neq c, \qquad f(x_n) = 0, \qquad x_n \to c.$$

(a) Prove that $f(c) = f'(c) = 0$.

(b) Suppose also that f is differentiable on I and $f''(c)$ exists. Prove that $f''(c) = 0$.

Question 11.

Let f be differentiable on an open interval containing $[a, b]$, where $a < b$. Suppose

$$f'(a) < \lambda < f'(b), \qquad \lambda \in \mathbb{R}.$$

Prove that there is a point $c \in (a, b)$ such that $f'(c) = \lambda$. Do not assume that f' is continuous.

3. Thomas Textbook Exercises

Use the same edition of Thomas' textbook adopted for this course.

Chapter 1. Limits and Differentiation

Section 1.2. Derivatives

§1.2.1. Derivatives of a function

Exercises 3.1: 18, 38. Exercises 3.2: 26, 54.

§1.2.2. Rules of derivatives and calculating

Exercises 3.3: 32, 56. Exercises 3.5: 46, 60.

Exercises 3.6: 56, 82. Exercises 3.7: 14, 38.